Eye Disease Classification (First Week)

Objective:

Build a deep learning classifier capable of diagnosing 8 retinal disease categories (Normal, Diabetic Retinopathy, Glaucoma, Cataract, AMD, Hypertension, Myopia, Others) from fundus images, using the unified dataset of 11,839 images aggregated.

Tasks

- Explore the dataset: total images per class, train/val/test splits, understand each eye disease class.
- Visualize sample fundus images from each of the 8 disease classes.
- Analyze class imbalance (Normal: 4698 vs Hypertension: 88) and decide on a handling strategy (class weighting, oversampling, etc.)
- Implement a custom CNN from scratch with a softmax output head for 8-class classification
- Train the scratch CNN and record accuracy, per-class precision, recall, and F1-score.
- Load a pretrained model (VGG or ResNet) and fine-tune it on the dataset.
- Compare both models and discuss the effect of class imbalance on per-class performance and which model is better for real-time deployment (note Consider the speed and the accuracy)

Completion Requirements

- A class distribution plot is produced showing all 8 classes with their sample counts.
- Class imbalance is explicitly addressed in the training setup (e.g., weighted loss)
- A CNN built from scratch achieves reasonable training convergence (loss decreases)
- A pretrained model is fine-tuned and outperforms the scratch CNN
- A per-class classification report (precision, recall, F1) is reported for both models
- Code is pushed to GitHub with ReadMe

Resources

- https://arxiv.org/abs/1409.1556
- https://arxiv.org/abs/1512.03385